Cycle Test 2: worked solutions

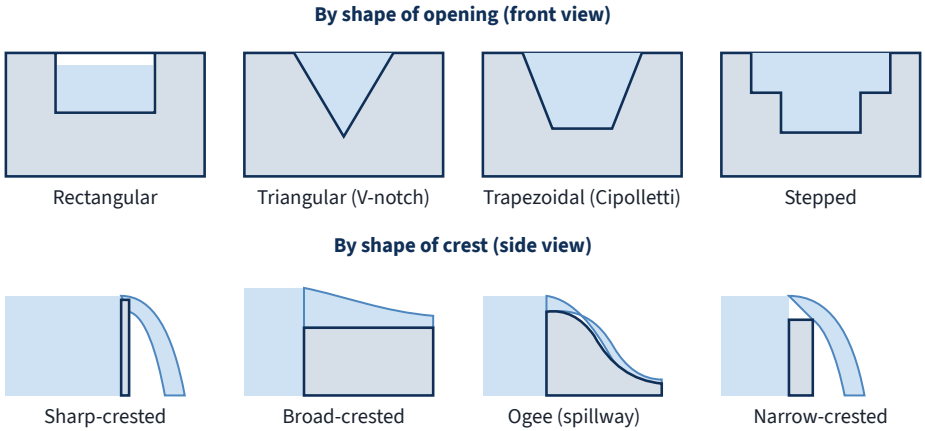
13 Oct 2025 · 20 marks · 1 hour · CEPC12

Part A (1 mark each)

Q1

1 mark

> Classify different types of weirs with neat sketches.



- **Shape of opening**: rectangular, triangular (V-notch), trapezoidal (Cipolletti), stepped.
- **Shape of crest**: sharp-crested, broad-crested (crest width > H/2), narrow-crested (width < H/2), ogee-shaped.
- **End contractions**: suppressed (crest spans the full channel width) or contracted.
- **Discharge condition**: free (nappe springs clear) or submerged/drowned (downstream level above crest).

Q2

1 mark

> What are major and minor losses?

- **Major loss**: loss of head due to **friction** along the length of a pipe, $h_f = 4fLV^2/(2gd)$ (Darcy–Weisbach). Large in long pipes.
- **Minor losses**: losses due to **local disturbances** in the flow: sudden enlargement $(V_1 - V_2)^2/2g$, sudden contraction, entrance $(0.5V^2/2g)$, exit $(V^2/2g)$, bends, fittings, valves and obstructions. Usually small compared with friction in long pipes.

Q3

1 mark

> Write short notes on the hydraulic coefficients of an orifice and show the relation among them.

- **C_c (coefficient of contraction)** = area of jet at vena contracta ÷ area of orifice (≈ 0.61–0.69).
- **C_v (coefficient of velocity)** = actual velocity at vena contracta ÷ theoretical velocity $\sqrt{(2gH)}$ (≈ 0.95–0.99).
- **C_d (coefficient of discharge)** = actual discharge ÷ theoretical discharge (≈ 0.61–0.65).
- **C_r (coefficient of resistance)** = loss of head in the orifice ÷ head of water.

Relation: $Q_{act} = (C_c\, a)(C_v\sqrt{(2gH)}) \Longrightarrow \boxed{C_d = C_c \times C_v}$

Q4

1 mark

> Define and explain the terms: (i) hydraulic gradient line (ii) total energy line.

- **HGL**: the line joining the points representing the **piezometric head ($p/\rho g + z$)** along a pipe; it is the level to which water would rise in piezometers fixed along the pipe.

- **TEL**: the line joining the points of **total head ($p/\rho g + V^2/2g + z$)**. It lies **$V^2/2g$ above the HGL** and always slopes downward in the direction of flow because of losses (drops sharply at minor losses).

Q5

1 mark

> Define the Hagen–Poiseuille equation.

For **laminar flow** in a circular pipe, the loss of head (pressure drop) over length L is

$$\boxed{h_f = (p_1 - p_2)/\rho g = 32\,\mu\,\bar{u}\,L / (\rho\, g\, D^2)}$$

where $\bar{u}$ = mean velocity and D = diameter. Equivalent to $f_{Darcy} = 64/Re$.

Part B (3 marks each)

Q6

3 marks

> A pipeline carrying oil (S = 0.87) changes in diameter from 200 mm at A to 500 mm at B, which is 4 m higher. Pressures at A and B are 9.81 N/cm² and 5.886 N/cm²; discharge 200 l/s. Find the loss of head and the direction of flow.

$\rho = 870$ kg/m³, $\rho g = 8534.7$ N/m³. $Q = 0.2$ m³/s. Take A as datum.

$V_A = 0.2/(\pi/4 \times 0.2^2) = 6.366$ m/s; $V_B = 0.2/(\pi/4 \times 0.5^2) = 1.019$ m/s

Total energy at A = $p_A/\rho g + V_A^2/2g + z_A = 98{,}100/8534.7 + 6.366^2/19.62 + 0 = 11.494 + 2.066 + 0 =$ **13.56 m**

Total energy at B = $58{,}860/8534.7 + 1.019^2/19.62 + 4 = 6.897 + 0.053 + 4 =$ **10.95 m**

$E_A > E_B$, so **flow is from A to B**, and the loss of head = 13.56 − 10.95 = **2.61 m (of oil)**.

Q7

3 marks

> Derive an expression for Bernoulli's equation from first principles and state the assumptions. Explain its practical applications with examples.

Euler's equation of motion along a streamline. Take a small cylindrical element of area dA and length ds along a streamline, inclined at θ to the horizontal. Forces in the s-direction:

- pressure: $p\,dA - (p + \partial p/\partial s\,ds)\,dA = -\partial p/\partial s\,ds\,dA$
- weight component: $-\rho g\,dA\,ds\cos\theta$ ($\cos\theta = dz/ds$)

Mass × acceleration: $\rho\,dA\,ds \times a_s$, with $a_s = V\,dV/ds$ for steady flow. Equating:

$$-(\partial p/\partial s)\,ds\,dA - \rho g\,dA\,ds\,(dz/ds) = \rho\,dA\,ds\,V\,(dV/ds)$$

$$\mathbf{dp/\rho + g\,dz + V\,dV = 0}\ \text{(Euler's equation)}$$

Integrating for an incompressible fluid (ρ constant):

$$p/\rho + gz + V^2/2 = \text{constant} \Longrightarrow \boxed{\mathbf{p/\rho g + V^2/2g + z = constant}}$$

i.e. pressure head + velocity head + datum head is constant along a streamline.

Assumptions: ideal (non-viscous) fluid; steady flow; incompressible; irrotational; flow along a streamline; only gravity and pressure forces act.

Applications:

- **Venturimeter and orifice meter**: flow measurement from the pressure drop at a constriction.
- **Pitot tube**: velocity of flow / aircraft speed from stagnation pressure.
- **Flow from tanks and orifices** (Torricelli: $V = \sqrt{(2gh)}$).
- **Siphons, nozzles, jets**; pipe network design (with loss terms added: $p_1/\rho g + V_1^2/2g + z_1 = p_2/\rho g + V_2^2/2g + z_2 + h_L$).
- Lift on aircraft wings, carburettors, spray guns.

Q8 (a)

1.5 marks

> A 100 mm pipe has a nozzle at its end with outlet diameter 50 mm. The discharge through the nozzle is 20 l/s and the pressure at the base of the nozzle is 5.886 N/cm^2. Find the coefficient of discharge (base and outlet at the same elevation).

Head at the base h = 58,860/9810 = 6.0 m (outlet at atmospheric pressure). a_1 = 0.007854 m^2, a_2 = 0.001963 m^2.

Bernoulli: $p_1/\rho g + V_1^2/2g = V_2^2/2g$, with $a_1V_1 = a_2V_2$

$Q_{th} = a_1a_2\sqrt{(2gh)} / \sqrt{(a_1^2 - a_2^2)} = (0.007854 \times 0.001963 \times \sqrt{(2 \times 9.81 \times 6)}) / 0.007605 =$ **0.0220 m^3/s**

$C_d = Q_{act}/Q_{th} = 0.020/0.0220 =$ **0.909**

Q8 (b)

1.5 marks

> Water flows in a rectangular channel 1 m wide and 0.75 m deep. Find the discharge over a rectangular weir of crest length 60 cm if the head over the crest is 20 cm and water from the channel flows over the weir. C_d = 0.62.

Without velocity of approach: $Q = (2/3)\, C_d\, L\, \sqrt{(2g)}\, H^{3/2} = (2/3)(0.62)(0.6)(4.429)(0.2)^{1.5} =$ **0.0983 m^3/s**

Velocity of approach $V_a = Q/(1 \times 0.75) = 0.131$ m/s → $h_a = V_a^2/2g = 0.000875$ m

$Q = (2/3)\, C_d\, L\, \sqrt{(2g)}\, [(H + h_a)^{3/2} - h_a^{3/2}] = 1.0985 \times [(0.200875)^{1.5} - (0.000875)^{1.5}]$

$Q =$ **0.0989 m^3/s ≈ 98.9 l/s**

Q9

3 marks

> A sudden contraction from 50 cm to 25 cm in a horizontal pipeline changes the pressure from 10,500 kgf/m^2 (103,005 N/m^2) to 6900 kgf/m^2 (67,689 N/m^2). Find the rate of flow, taking C_c = 0.65. If there is then a sudden enlargement from 25 cm to 50 cm and the pressure at the 25 cm section is 6900 kgf/m^2, what is the pressure at the 50 cm section?

Contraction (1 → 2). $V_1 = V_2/4$ (area ratio 4). Loss $h_c = (1/C_c - 1)^2 V_2^2/2g = (1/0.65 - 1)^2 V_2^2/2g = 0.2899\, V_2^2/2g$.

$(p_1 - p_2)/\rho g = 35{,}316/9810 = 3.6$ m

$3.6 = V_2^2/2g - V_1^2/2g + h_c = (V_2^2/2g)(1 - 1/16 + 0.2899) = 1.2274\, V_2^2/2g$

$V_2^2/2g = 2.933$ m → $V_2 = 7.586$ m/s

$Q = \pi/4 \times 0.25^2 \times 7.586 =$ **$0.372\ m^3/s$**

Enlargement (3 → 4). $V_3 = 7.586$ m/s, $V_4 = 1.896$ m/s; loss $h_e = (V_3 - V_4)^2/2g$.

$p_4/\rho g = p_3/\rho g + V_3^2/2g - V_4^2/2g - (V_3 - V_4)^2/2g$

$= 6.90 + 2.933 - 0.183 - (5.689)^2/19.62 = 6.90 + 2.933 - 0.183 - 1.650 =$ **8.00 m**

$p_4 =$ **8000 kgf/m² = 78,480 N/m²**

Q10

3 marks

> The difference in water levels in two tanks connected by three pipes in series (lengths 300, 170, 210 m; diameters 300, 200, 400 mm) is 12 m. Find the rate of flow if f = 0.005, 0.0052 and 0.0048, (i) considering minor losses, (ii) neglecting minor losses.

Continuity with $V_1 = V$: $V_2 = V(0.3/0.2)^2 = 2.25V$; $V_3 = V(0.3/0.4)^2 = 0.5625V$. Friction loss $h_f = 4fLV^2/(2gd)$. All terms are written as multiples of $V^2/2g$.

Loss	Expression	Coefficient × $V^2/2g$
Friction, pipe 1	$4(0.005)(300)/0.3 \times 1$	20.00
Friction, pipe 2	$4(0.0052)(170)/0.2 \times 2.25^2$	89.51
Friction, pipe 3	$4(0.0048)(210)/0.4 \times 0.5625^2$	3.19
Entrance	$0.5\, V_1^2/2g$	0.50
Contraction 1→2	$0.5\, V_2^2/2g$	2.53
Enlargement 2→3	$(V_2 - V_3)^2/2g = (1.6875)^2$	2.85
Exit	$V_3^2/2g$	0.32

(i) With minor losses: $12 = 118.89\, V^2/2g$ → $V = \sqrt{(2 \times 9.81 \times 0.1009)} = 1.407$ m/s

$Q = \pi/4 \times 0.3^2 \times 1.407 =$ **$0.0995\ m^3/s$**

(ii) Neglecting minor losses: $12 = 112.69\ V^2/2g \rightarrow V = 1.445$ m/s

$Q = \pi/4 \times 0.3^2 \times 1.445 =$ **$0.1022\ m^3/s$**